

Technical Document #577: Cloud Computing - Comprehensive Overview

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Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Platform as a Service (PaaS) provides a platform for developing and deploying applications. Heroku and Google App Engine are PaaS examples.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Serverless computing allows developers to run code without managing servers.
- Platform as a Service (PaaS) provides a platform for developing and deploying applications.
- Containerization packages applications with their dependencies.